

Favard length of non-homogeneous random Cantor sets

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Favard length problem

Favard length

Let $\pi_\theta(x) := x \cdot (\cos \theta, \sin \theta)$.

Favard length (or **Buffon's needle probability**) of a Borel set $K \subset \mathbb{R}^2$ is

$$\text{Fav}(K) := \int_0^\pi |\pi_\theta(K)| d\theta,$$

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$$\dim K > 1 \quad \Rightarrow \quad \text{Fav}(K(r)) \gtrsim 1$$

$$\dim K = s < 1 \quad \Rightarrow \quad \text{Fav}(K(r)) \sim r^{1-s}$$

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Theorem (Besicovitch 1939)

Let $K \subset \mathbb{R}^2$ with $0 < \mathcal{H}^1(K) < \infty$.

$$\text{Fav}(K) = 0 \quad \Leftrightarrow \quad K \text{ is purely unrectifiable,}$$

i.e. $\mathcal{H}^1(K \cap \Gamma) = 0$ for any rectifiable curve Γ .

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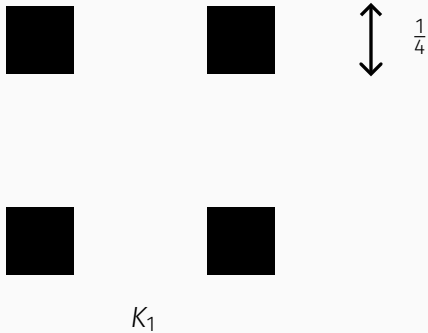
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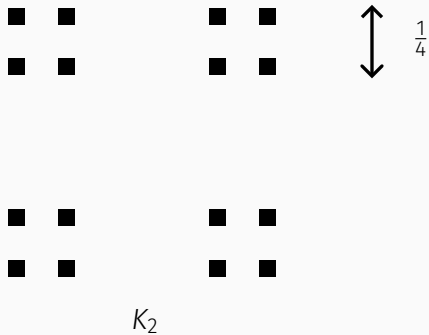
i.e. $\mathcal{H}^1(K \cap \Gamma) = 0$ for any rectifiable curve Γ .

In particular: $\text{Fav}(K) = 0$ for nice self-similar sets of dimension 1.

Four-corners Cantor set



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K_3

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Since $\text{Fav}(K) = 0$, we get

$$\text{Fav}(K_n) \xrightarrow{n \rightarrow \infty} 0.$$

Favard length problem

Problem (Peres-Solomyak '02)

What is the rate of decay of $\text{Fav}(K_n) \rightarrow 0$?

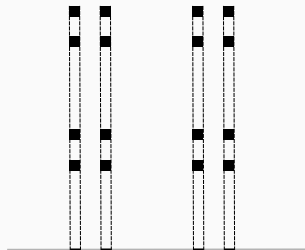
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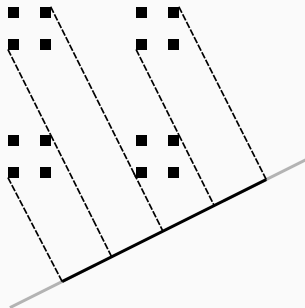
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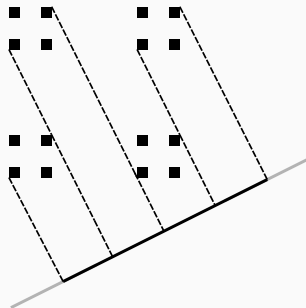
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What happens on average???

Lower bounds

Theorem (Mattila '90)

For any compact $E \subset \mathbb{R}^2$

$$\text{Fav}(E) \gtrsim \text{Cap}_1(E),$$

where

$$\text{Cap}_1(E) = \sup_{\mu, \text{supp } \mu \subset E} \frac{1}{\mathcal{I}_1(\mu)}.$$

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Theorem (Bateman-Volberg '08)

This is not sharp:

$$\text{Fav}(K_n) \gtrsim \frac{\log n}{n}.$$

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- $\text{Fav}(E_n) \lesssim n^{-c}$ for some other self-similar sets [Bond-Volberg '10], [Łaba-Zhai '10], [Bond-Łaba-Volberg '11], [Łaba-Marshall '22], [Marshall '26+]

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- proofs: Fourier analysis, estimates for trigonometric polynomials

Motivation - Vitushkin's conjecture

Vitushkin's conjecture (1967) on removable sets for bounded analytic functions predicts

$$\gamma(E) \geq c \operatorname{Fav}(E) \quad \forall \text{ compact } E \subset \mathbb{R}^2,$$

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We don't even know that for the four-corners Cantor set!

$$\operatorname{Fav}(K_n) \lesssim_{\varepsilon} n^{-1/5+\varepsilon} \quad (\text{Marshall '26+})$$

Random Cantor sets

- deterministic sets: unsatisfactory Favard length estimates 😞

Random Cantor sets

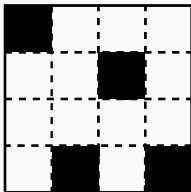
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- many **random** Cantor sets: sharp Favard length estimates! 😊

Theorem (Peres-Solomyak '02, Chang-Shmerkin-Suomala '25)

For many 1-dimensional grid-based random Cantor sets

$$\text{Fav}(E_n) \sim \frac{1}{n}$$

almost surely and in expectation.



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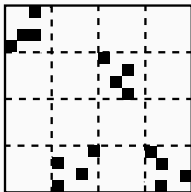
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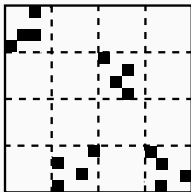
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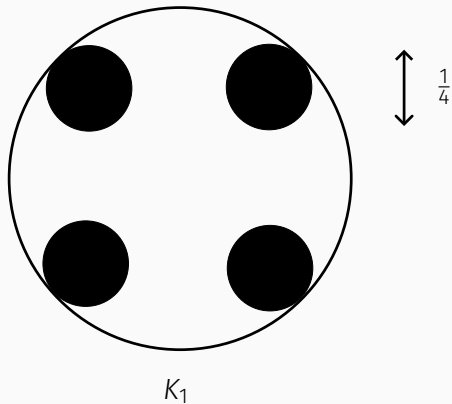
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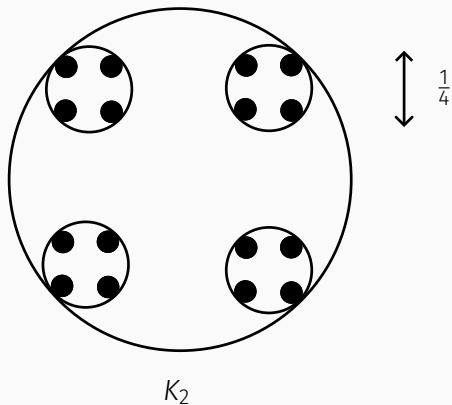


$\rightsquigarrow$ for such E_n Mattila's bound is sharp a.s.

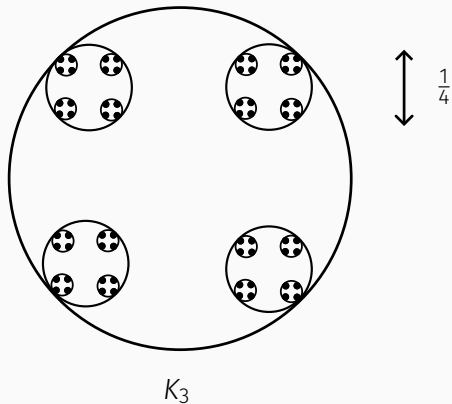
Four-corners Cantor set revisited



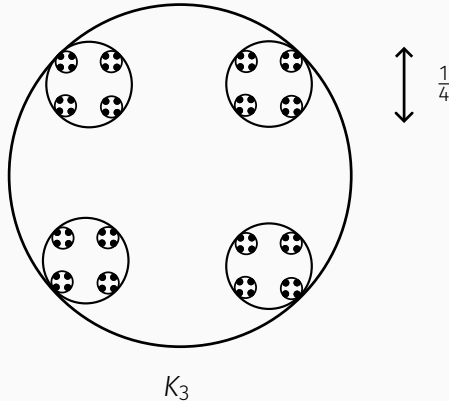
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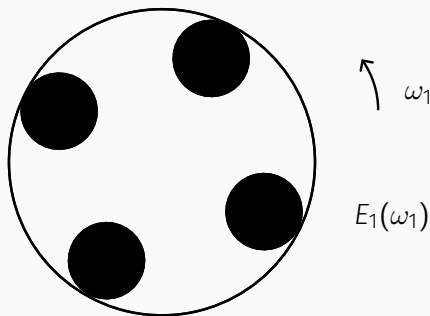


$K = \bigcap_n K_n$ – usual four corners Cantor set

Random Cantor sets – disc models

◦ “synchronized rotations model”

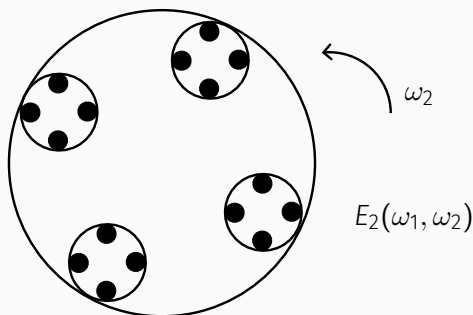
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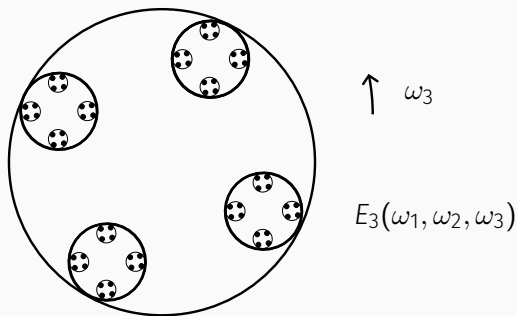
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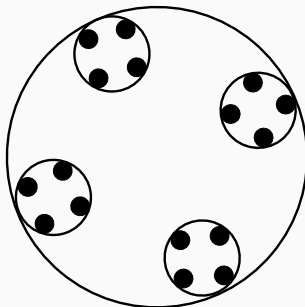
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- “independent rotations model”

$\rightsquigarrow$ i.i.d. random variable ω_D for each disc D



Theorem (Zhang '20)

For the synchronized model

$$\mathbb{E} \text{Fav}(E_n) \sim \frac{1}{n}.$$

Theorem (Vardakis-Volberg '24)

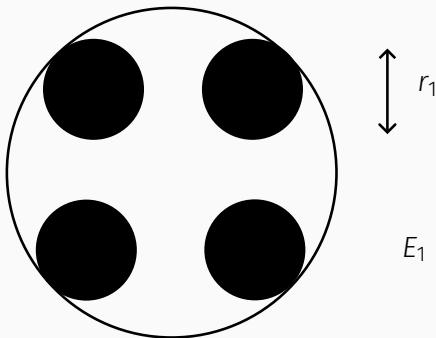
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Non-homogeneous Cantor sets

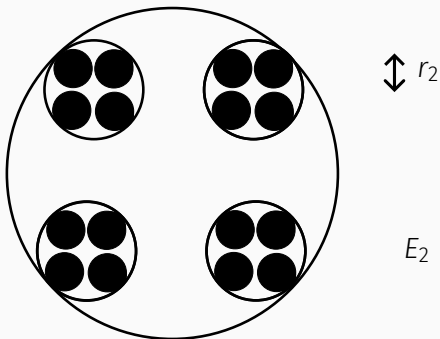
Non-homogeneous four-corners set (Garnett 70s)

For $r_0 > r_1 > r_2 > \cdots > 0$ satisfying $r_{n+1} \leq r_n/3$ consider:



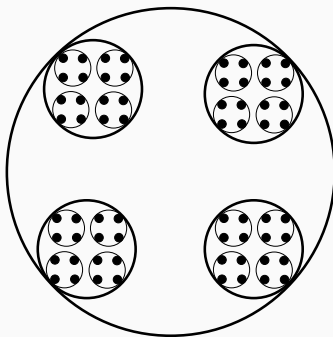
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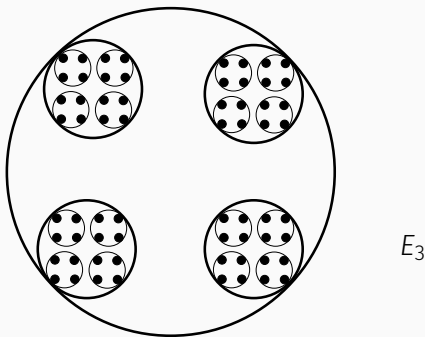
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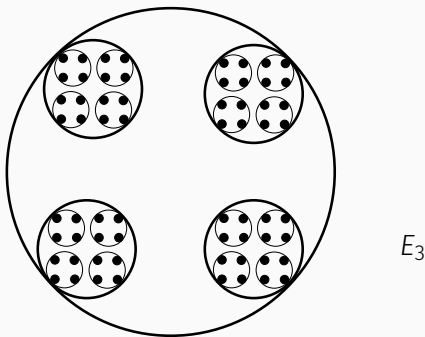
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- $E = \bigcap_n E_n$ - non-homogeneous four-corners Cantor set
- $r_n = 4^{-k} \rightsquigarrow$ usual four-corners Cantor set

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Problem (Mattila '04)

Characterize sequences $(r_n)_{n \in \mathbb{N}}$ such that $\text{Fav}(E) = 0$.

What is known

Sufficient conditions:

- Besicovitch projection theorem:

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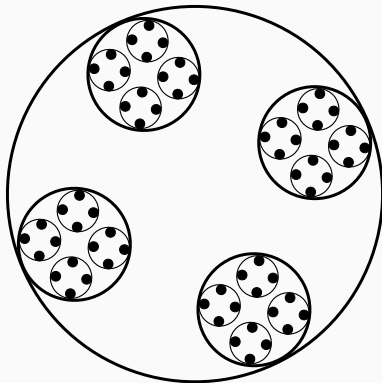
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Random variant

[Chang-D.-Del Nin '26] studies a random variant of these sets:

- “synchronized rotations model” $\rightsquigarrow (\omega_n)_{n \geq 1}$ - i.i.d. uniform r.v. on $[0, 2\pi]$
- “independent rotations model” $\rightsquigarrow$ i.i.d. uniform r.v. ω_D for each disc D



Solution for the random variant

Theorem (Chang-D.-Del Nin '26)

For both models

$$\mathbb{E} \text{Fav}(E_n) \sim \text{Cap}_1(E_n) \sim \left(\sum_{k=0}^n \frac{1}{r_k 4^k} \right)^{-1}. \quad (1)$$

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- in the independent rotations model we adapt [CSS '25] to get an a.s. version of (1).

Proof for synchronized rotations

Setup

- Fix $r_0 > r_1 > r_2 > \dots$
- Goal:

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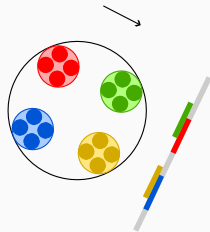
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Setup

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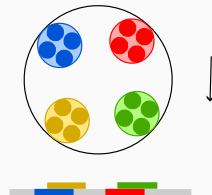
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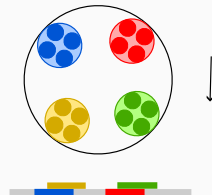
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- $a_n = \mathbb{E}|\pi_0(E_n)|$
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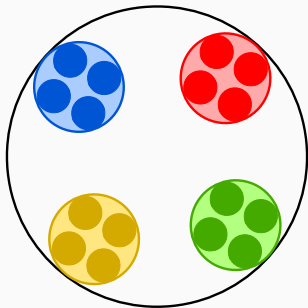
Focus on the homogeneous case $r_n = 4^{-n}$:

$$\boxed{a_n \leq a_{n-1} - c a_{n-1}^2} \rightsquigarrow \mathbb{E} \text{Fav}(E_n) \sim a_n \lesssim \frac{1}{n}$$

Proof of inductive inequality (Zhang '20)

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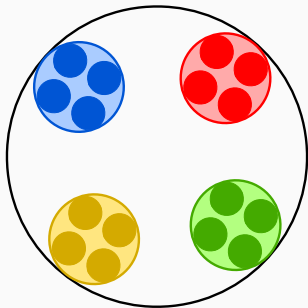
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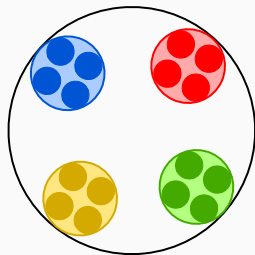
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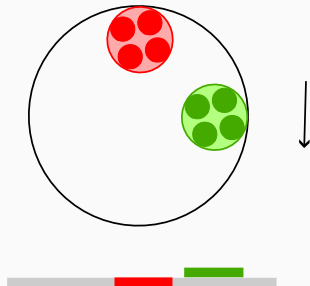
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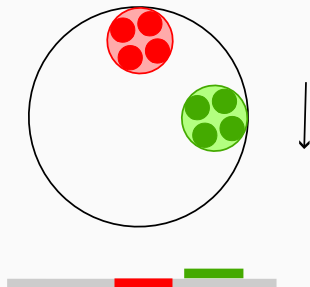


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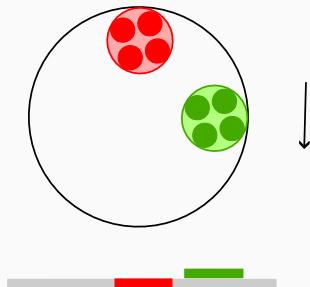
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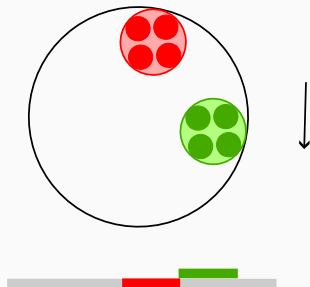
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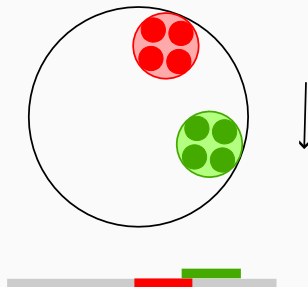
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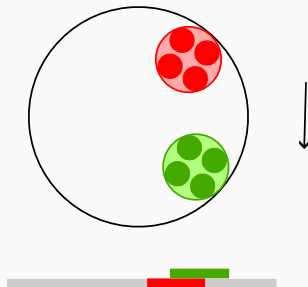
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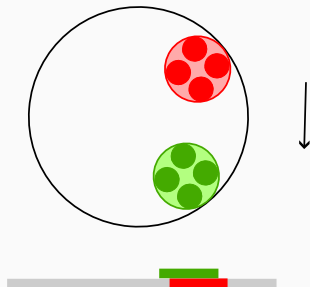
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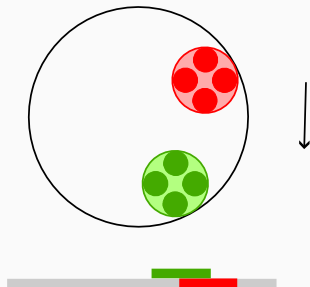
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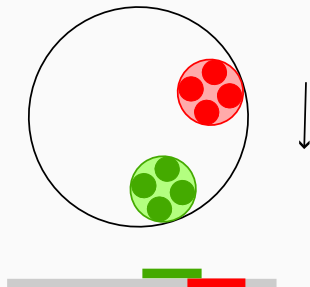
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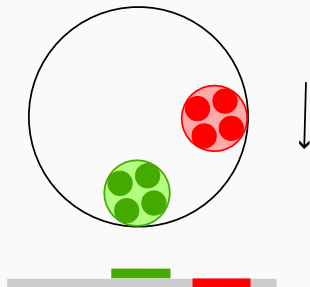
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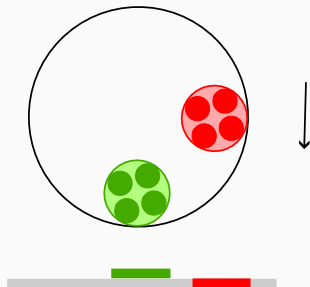
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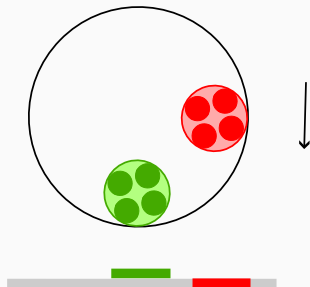
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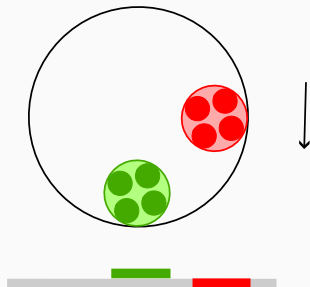
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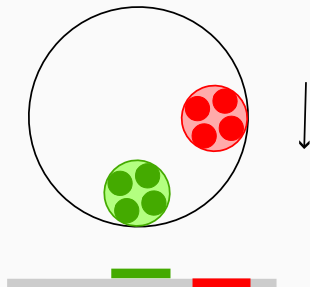
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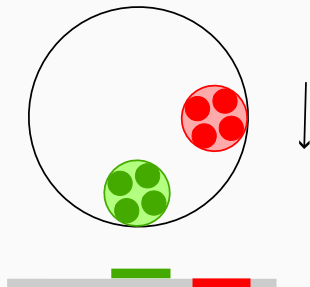
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Fix $\omega_2, \dots, \omega_n$, set $F := \pi_0(\frac{1}{4}E_{n-1}(\omega_2, \dots, \omega_n))$.

$$\pi_0(S_j) = F + \pi_0(v_j(\omega_1)) = F + \frac{3}{4} \cos(j\pi/2 + \omega_1).$$

Compute $\mathbb{E}_{\omega_1}$:

$$\begin{aligned} \mathbb{E}_{\omega_1} |\pi_0(S_1) \cap \pi_0(S_2)| &= \int_0^{2\pi} |F \cap (F + \frac{3}{4} \cos(\pi/2 + \omega_1) - \frac{3}{4} \cos(\pi + \omega_1))| d\theta \\ &\approx \int_{\mathbb{R}} |F \cap (F + t)| dt = \iint \mathbf{1}_F(s) \mathbf{1}_{F+t}(s) ds dt = \iint \mathbf{1}_F(s) \mathbf{1}_{F-s}(t) ds dt = |F|^2 \\ &= |\pi_0(\frac{1}{4}E_{n-1}(\omega_2, \dots, \omega_n))|^2 \end{aligned}$$



Size of the intersection

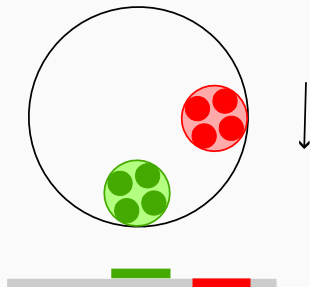
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Thank you!